

**Econ 272 / GSB 607**  
**Spring 2025**

**Department of Economics**  
**Stanford University**

**PROBLEM SET IV**

**DUE: SUNDAY APRIL 27<sup>th</sup>, 2025, 6PM.**

1. Theoretical Calculations

- (a) Consider the figure on the left of page 16 of the slides, drawn from Pearl 1995. Is there a set of observed variables we can condition on to use the backdoor criterion to identify the effect of fumigation on yields?
- (b) Give an economic example of M-bias where conditioning on a proper pre-treatment variable would introduce bias that was not present if you did not condition on this pre-treatment variable. Make the example as specific as possible.
- (c) Grilliches and Mason (Griliches, Z., & Mason, W. M. (1972). Education, income, and ability. *Journal of political Economy*, 80(3, Part 2), S74-S103.) are interested in estimating the causal effect of education on earnings. They use information on family background, education before and education after military service, length of military service, AFQT test (Armed Forces Qualification Test, taken prior to military service), and earnings for a sample of veterans. Build a DAG for the variables family background, education before military service, education after military service, earnings, length of military service, AFQT score. Grilliches and Mason come up with their own graph, so don't just copy that, but think about which links you think should be there, and which should not. Is there a backdoor strategy that identifies the effect of education before military service on earnings? How about the effect of the total level of education on earnings?